R Code Execution Outputs

```
vec <- c(6,7,9,3,5,1)
sorted_vec <- sort(vec,decreasing = TRUE)

vec[2]
# [1] 7

vec[c(2,4)]=c(44,99)

sq <- seq(2,10,2)

sq
# [1] 2 4 6 8 10
```

```
sq <- seq(105,150,3)

sq
# [1] 105 108 111 114 117 120 123 126 129 132 135 138 141 144 147 150
```

```
sq <- seq(105,111,3)
ss2 <- seq(113,119,3)
sq3 <- c(sq,ss2)

sq3
# [1] 105 108 111 113 116 119
```

```
repi <- rep(100,times=5)

repi
# [1] 100 100 100 100 100
```

```
task <- c(1,rep(2,2),rep(3,3),rep(4,4),rep(5,5))

task
# [1] 1 2 2 3 3 3 4 4 4 4 5 5 5 5 5
```

```
tp <- seq(1,5,1)

tp
# [1] 1 2 3 4 5

tp1 <- rep(tp,tp)

tp1
# [1] 1 2 2 3 3 3 4 4 4 4 5 5 5 5 5

rep(1:5,1:5)
# [1] 1 2 2 3 3 3 4 4 4 4 5 5 5 5 5
```

```r
mat <- matrix(1:9,nrow=3,ncol=3)

mat
# [,1] [,2] [,3]
# [1,] 1 4 7
# [2,] 2 5 8
# [3,] 3 6 9
```

```r
mat <- matrix(1:9,nrow=3,ncol=3,byrow=TRUE)

mat
# [,1] [,2] [,3]
# [1,] 1 2 3
# [2,] 4 5 6
# [3,] 7 8 9
```

```r
mat <- matrix(seq(11,27,2),nrow=3,ncol=3,byrow=TRUE)

mat
# [,1] [,2] [,3]
# [1,] 11 13 15
# [2,] 17 19 21
# [3,] 23 25 27
```

```r
rownames(mat) <- c("row1","row2","row3")
colnames(mat) <- c("col1","col2","col3")

mat
# col1 col2 col3
# row1 11 13 15
# row2 17 19 21
# row3 23 25 27
```

```r
print(mat[2,2])
# [1] 19

print(mat[c(2,3),c(2,3)])
# col2 col3
# row2 19 21
# row3 25 27

mat[2:3,2:3]
# col2 col3
# row2 19 21
# row3 25 27
```

```r
mat[3,]
# col1 col2 col3
# 23 25 27

mat[,3]
# row1 row2 row3
# 15 21 27
```

```r
mat1 <- matrix(1:4,nrow=2)
mat2 <- matrix(5:8,nrow=2)
mat3 <- mat1+mat2

mat3
# [,1] [,2]
# [1,] 6 10
# [2,] 8 12
```

```r
mat3 <- mat1*mat2

mat3
# [,1] [,2]
# [1,] 5 21
# [2,] 12 32
```

```r
mat3 <- mat1%*%mat2

mat3
# [,1] [,2]
# [1,] 23 31
# [2,] 34 46
```

```r
mat3 <- t(mat3)

mat3
# [,1] [,2]
# [1,] 23 34
# [2,] 31 46
```

```r
mat_inverse <- solve(mat3)

mat_inverse
# [,1] [,2]
# [1,] 11.50 -8.50
# [2,] -7.75 5.75
```